

Problem-Solution Fit

Date	02 October 2026
Team ID	SWTID-2026-5625
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS] • Beginners and busy students/professionals (18–40) who want to get fit • People with minor health concerns needing safer workouts • Users who can't afford a trainer or dietitian	6. CUSTOMER LIMITATIONS [CL] - Eg. budget, devices • Limited budget and no gym equipment • Little fitness or nutrition knowledge • Phone/laptop with internet only; short on time	5. AVAILABLE SOLUTIONS [AS] Pros & cons • Personal trainers: personalized but expensive • YouTube/blogs: free but generic, unverified • Fitness apps: feature-rich but paid, often not health-aware • DIY plans: cheap but time-consuming, error-prone
2. PROBLEMS / PAINS + FREQUENCY [PR] • Generic plans ignore age, weight, goal, health (daily/weekly) • Hard to plan workouts and meals together (weekly) • Plans don't adapt to progress (ongoing)	9. PROBLEM ROOT / CAUSE [RC] • Lack of affordable, personalized guidance • Plans ignore individual health and goals • No easy feedback loop to adjust plans	7. BEHAVIOR + ITS INTENSITY [BE] • Searches online, tries random routines (high at first) • Starts strong, quits within weeks (high drop-off) • Rarely tracks progress or updates plan
3. TRIGGERS TO ACT [TR] • New Year/health goals, weight gain, doctor's advice • Upcoming events (wedding, trip, sports trial) • Social media or friends' transformations	10. YOUR SOLUTION [SL] • FitBuddy is a FastAPI web app using Google Gemini AI. From name, age, weight, goal, intensity and optional health info it generates: • Personalized 7-day workout plan • 7-day food plan • Nutrition and recovery tips • Health-aware workout considerations • Feedback-based plan updates (SQLite/SQLAlchemy)	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Google, YouTube, Instagram, fitness apps, web apps OFFLINE • Gyms, trainers, friends/family advice, doctors
4. EMOTIONS BEFORE / AFTER [EM] • Before: confused, unmotivated, overwhelmed, afraid of injury • After: confident, guided, motivated, in control		

Team Members and Roles

Sharmila D – Team Lead	Managed overall development; handled prerequisites; core FitBuddy functionality; team coordination; local deployment.
Bhavana A – Backend Developer	Project workflow; FastAPI backend; routing and user-input handling; testing during local deployment.
Rajashree K – AI/GenAI Developer	Gemini AI integration; Gemini model research; main application logic; conclusion and AI functionality.
Siva Sundari V – Frontend/UI Developer	Application architecture; UI design; frontend components; simple, user-friendly interface.
Praveena K – Development and Template Developer	Environment setup; dynamic FastAPI + Jinja2 templates; connecting templates to backend; development support and testing.